
Homework 1

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MA 564 Section A1

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Problem 1

By definition, $\emptyset, \mathcal{T} \in \mathbb{R}$. Now let $\{(-\infty, p)\}_{p \in P} \subseteq \mathcal{T}$, and consider the union over this collection. Let $a = \sup P$, and let $x \in (-\infty, a)$.

Problem 2

Problem 3

Note that $X^c = X \setminus X = \emptyset$, and $\emptyset^c = X \setminus \emptyset = X$. Therefore, $\emptyset, X$ are closed.

Let $\{X_a\}_{a \in A}$ be a collection of closed sets in X . It follows that there exists a collection $\{Y_a\}_{a \in A}$ of open sets such that $Y_a^c = X_a$. It follows that

$$\bigcap_{a \in A} X_a = \bigcap_{a \in A} Y_a^c.$$

By De Morgan's laws, this equals

$$\left(\bigcup_{a \in A} Y_a \right)^c.$$

By the definition of a topology, $\bigcup_{a \in A} Y_a$ is open, so $\left(\bigcup_{a \in A} Y_a \right)^c$ is closed.

Let $\{X_k\}_{k=1}^k$ be a finite collection of closed sets in X . It follows that there exists a collection $\{Y_i\}_{i=1}^k$ of open sets such that $Y_i^c = X_i$ for all i . By De Morgan's laws, it follows that

$$\bigcup_{i=1}^k X_i = \bigcup_{i=1}^k Y_i^c = \left(\bigcap_{i=1}^k Y_i \right)^c.$$

By definition of a topology, since Y_i is open for all i , it follows that $\bigcap_{i=1}^k Y_i$ is open. Therefore, $\left(\bigcap_{i=1}^k Y_i \right)^c$ is closed.

Problem 4

Problem 5

Let X be Hausdorff, and consider the intersection

$$U := \bigcap_V \overline{V}$$

over all neighborhoods V of x . Note that $V \subseteq \overline{V}$, and by definition of an intersection, $U \subseteq \overline{V}$ for all neighborhoods V of X . Since $x \in V$ for all V , and $V \subseteq \overline{V}$, we have that $x \in U$. Now suppose

for contradiction that there exists $y \in U$ such that $y \neq x$. Since X is Hausdorff, there exist disjoint neighborhoods V_x, V_y of x, y , respectively. Therefore, $y \notin V_x$, and we must have $y \in \overline{V_x} \setminus V_x$. Idk how to finish the proof, but we're going to need to choose a third point.

Now suppose the aforementioned intersection $U = \{x\}$.

Problem 6

Problem 7